

BHANU PRAKASH

MLOps & AI Infrastructure Engineer · AWS · Kubernetes · Terraform · ML Platform Engineering

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PROFESSIONAL SUMMARY

MLOps and AI Infrastructure Engineer with 4+ years of experience building the cloud platforms, Kubernetes infrastructure, and automated pipelines that power data-intensive and AI-driven workloads at scale. Expert in designing GPU-ready container orchestration, scalable ML pipeline infrastructure, feature store integrations, and high-throughput data platforms on AWS and GCP. Proven record delivering 40% faster release cycles, 35% scalability improvements, and 25% cloud cost reductions in regulated FinTech and SaaS environments. Passionate about bridging the gap between ML teams and production infrastructure — getting models from notebook to production, reliably and fast.

TECHNICAL SKILLS

ML & AI Infrastructure: Kubeflow, MLflow (model registry & tracking), Ray, Argo Workflows, Jupyter Hub, GPU workload scheduling (NVIDIA), Feature Stores

Cloud Platforms: AWS (EC2, S3, SageMaker-ready, EKS, Lambda, SQS, RDS, IAM, CloudWatch), GCP (GKE, Vertex AI-ready, Compute Engine), Multi-Cloud

Container & Orchestration: Docker, Kubernetes, EKS, GKE, Helm, FluxCD, Argo CD, Karpenter, GPU node pools, Service Mesh (Istio)

ML Pipeline & Data: Argo Workflows, Apache Kafka (event streaming), S3 data lakes, Aurora/RDS, real-time inference pipelines, batch processing

Infrastructure as Code: Terraform, Terragrunt, AWS CloudFormation, Pulumi, Ansible

CI/CD & GitOps: GitHub Actions, GitLab CI/CD, Jenkins, Argo CD, FluxCD, DORA Metrics, model deployment automation

Observability: Prometheus, Grafana, ELK Stack, OpenTelemetry, CloudWatch, AWS X-Ray — including ML model monitoring & drift detection

Security & Compliance: DevSecOps, HashiCorp Vault, IAM, Zero Trust, Policy-as-Code, PCI-DSS, SOC 2, FedRAMP

Languages: Python, Bash, Go, PowerShell

FinOps & Reliability: GPU/CPU right-sizing, Autoscaling, Cost Optimization, SLIs/SLOs, MTTR Reduction, Incident Management

CERTIFICATIONS

AWS Certified Solutions Architect · AWS Certified DevOps Engineer – Professional · CKA – Certified Kubernetes Administrator [Update as applicable]

PROFESSIONAL EXPERIENCE

Senior MLOps & DevOps Engineer · EZCORP · Irving, TX (Hybrid)

Jan 2025 – Present

- ◆ **AI/ML platform infrastructure:** Architected Kubernetes clusters (EKS, GKE, hybrid on-prem) optimized for containerized AI/ML workloads, real-time inference services, and data-intensive analytics pipelines processing millions of events daily.
- ◆ **ML pipeline automation:** Designed and deployed Argo Workflows-based ML pipeline infrastructure, enabling automated model training runs, validation gates, and reproducible deployment workflows across environments.
- ◆ **Scalable model serving:** Built and maintained high-availability microservices infrastructure supporting real-time prediction APIs and risk analytics models, achieving 99.99% uptime under production financial workloads.
- ◆ **Feature & data platform:** Partnered with data science teams to optimize containerized analytics and event streaming platforms, improving resource utilization and throughput for AI-driven financial risk modeling.
- ◆ **IaC for ML environments:** Built reusable Terraform/Terragrunt modules for GPU-ready, ML-optimized cloud environments — cutting provisioning time by 45% and eliminating config drift across teams.
- ◆ **MLOps observability:** Enhanced Prometheus/Grafana/ELK monitoring stack with ML-specific dashboards and alerting, reducing MTTR from ~45 min to ~12 min for production AI services.
- ◆ **Cost-optimized compute:** Implemented Karpenter-based autoscaling for dynamic ML workloads, reducing monthly cloud spend by ~25% while maintaining model inference throughput under peak load.
- ◆ **Compliance-ready AI delivery:** Integrated security scanning, policy-as-code, and compliance checks (PCI-DSS, SOC 2) into CI/CD pipelines, ensuring regulated AI and analytics deployments meet audit requirements.

DevOps Engineer – Cloud Infrastructure & AI Data Platforms · S&P Global · India (Remote) Nov 2021 – Jul 2023

- ◆ **AI data platform infrastructure:** Designed and delivered AWS infrastructure supporting large-scale financial data processing and market intelligence platforms, enabling analytics and AI-driven risk applications for global enterprise clients.

- ◆ **Pipeline acceleration:** Optimized CI/CD pipelines (GitHub Actions, Jenkins, AWS CodePipeline), cutting deployment time by 40% and accelerating delivery of data and ML-backed analytics applications.
- ◆ **Scalable ML-ready containers:** Led containerization strategy using Docker and Kubernetes (EKS/ECS), improving platform scalability by 35% and enabling data science teams to deploy and iterate models faster.
- ◆ **Reusable infrastructure:** Developed Terraform modules for ML-ready, audit-ready cloud environments — reducing provisioning time by 45% across multiple data and analytics platforms.
- ◆ **Observability for data workloads:** Built enterprise monitoring (CloudWatch, X-Ray, centralized logging), reducing MTTR by 30% across data pipelines and analytics services.
- ◆ **Compliance as code:** Implemented SOC 2, PCI-DSS, and FedRAMP controls directly in CI/CD pipelines — enabling continuous compliance for regulated AI and financial data platforms.

Infrastructure Platform Engineer – AI-Driven SaaS · Zenoti · India

Jan 2020 – Oct 2021

- ◆ **AI-ready platform infrastructure:** Designed and operated multi-tenant AWS/GCP Kubernetes infrastructure supporting data-intensive SaaS features including AI-driven scheduling, recommendations, and analytics.
- ◆ **Workload optimization for ML:** Tuned Kubernetes workload scheduling, cluster resource allocation, and service isolation strategies to support data science and AI-driven platform features efficiently.
- ◆ **GitOps for ML delivery:** Implemented FluxCD and Helm-based GitOps pipelines enabling automated, reproducible deployment of AI and analytics services across staging and production environments.
- ◆ **Platform automation:** Built Python, Bash, and Go tooling reducing manual operational overhead by ~50%, freeing data and ML engineers to focus on model development rather than infrastructure.
- ◆ **Secrets & security:** Deployed HashiCorp Vault-based secrets management and zero-trust access controls — protecting multi-tenant environments including sensitive AI model artifacts and training data.
- ◆ **Reliability for AI services:** Contributed to disaster recovery architecture and backup automation, achieving sub-30-minute RTO for uninterrupted AI-powered SaaS delivery.

EDUCATION

Master of Science, Computer Science · Texas A&M University, TX

GPA: 3.7 / 4.0

Bachelor of Engineering, Computer Science · Jain University, Bangalore, India

CGPA: 3.6 / 4.0